
UNIT 1 BASIC MEASUREMENT OF MORTALITY AND MORBIDITY

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1.1 INTRODUCTION

We studied in the previous block that mortality is one of the components of population change. Mortality means the departure of member from the households permanently due to death. The idea of mortality measurement is the utilization for the construction of life table which indicates the expectancy of life at different ages or the proportion of the population at one age that survives to another age.

Mortality is defined as the demographic event of death. Since death is a biological phenomenon that occurs just once to each individual, the analysis is simpler than, say, the study of fertility wherein the event of birth can occur with varying frequency among women. Mortality analysis begins with good quality data on deaths and population. These data are conventionally obtained from vital registration systems and population censuses respectively. The crude death rate and the specific death rates (age, sex, age-sex- cause of death specific) are simple measures of mortality. The other measures are based on the life tables. Since, death is biological phenomenon which happened once to each individual or unit of the society.

After reading this unit learner will be able to:

- (i) Explain the concept of mortality
- (ii) Discuss about the morbidity trends
- (iii) Describe the various methods or techniques of mortality analysis
- (iv) Narrate the use of mortality analysis

1.2 CONCEPT OF MORTALITY AND MORBIDITY

Some literature review is required before going to read the mortality and morbidity. Both mortality and morbidity are interlinked, but both are separate events. Morbidity is the condition of health, while mortality is the risk of dying due to various causes. A state of well being contributes to raise national productivity and improves many social indicators of better living. Health of an individual is defined as a state of complete physical, mental and social well being that enables an individual to be productive, free from disease and thus able to access the opportunities available in the environment (WHO, 1946). Morbidity led indoor treatment and “Hospitalization is a two-edged sword. When, patients with border line personality disorder were admitted. They enter an environment that reinforces the very behaviours that therapy is trying to extinguish” (Paris, 2002).

Health and illness are the most important issue in the discussion about hospitalization. A layman often says that illness occurs due to curse of God and good health is the blessing of the God. They also believe that cause of illness is due to over diet and under diet. There are two types of nutritional disorders, i.e. over nutrition and under nutrition. Both conditions show the path of hospital. In higher social-economic groups family members suffer from over nutrition. Dr. Swaminathan (an agricultural scientist) and Dr. Ramchandran a nutritionist expressed concern over malnutrition and its results. Ogden (1997) explain the eating behaviour in the form of theories that regarded the individual as passively external food cues and later theories emphasized the processing that food intake resulted from an interaction between individual and their environment. And recent theories describe eating in terms of self-control, self-efficacy and self-regulation. This is particularly salient in theories of overeating in normal weight individuals and eating disorders.

According to NSSO the definition of ailment (illness or injury) means any deviation from the state of physical and mental well being. An ailment may not cause any necessity of hospitalization. An ailing member is a normal member of the household who was suffering from any ailment during the reference period. In the survey, one will be treated as sick if one feels sick. This was also include among other things

such as cases of visual, hearing, speech, locomotor and mental disabilities; injuries were cover all types of damages, such as cuts, wounds, haemorrhage, fractures and burns caused by an accident, including bites to any part of the body; cases of abortion - natural or accidental. In the survey cases of sterilization, insertion of IUD, getting MTP etc; cases of pregnancy and childbirth were not included. Thus, mortality is the rate of death while morbidity is the state of illness. Mortality may be due to illness whereas morbidity due to various causes either diseases and diseases may be communicable or non communicable.

1.3 DATA FOR MORTALITY AND MORBIDITY MEASURES

Sample Registration System (SRS) which is called dual record system of registration and Civil Registration System (CRS) which is called vital statistics data are best sources of mortality analysis on annual basis. SRS data is available at national and state level while CRS data has covered at district level. Quality of CRS data is not upto mark or standards. Some states have reported bad quality of CRS data. Government of India has also conducted Annual Health Survey (AHS) in 9 states BIMARU states (Bihar, Madhya Pradesh, Rajasthan and Uttar Pradesh, term BIMARU has coined by Ashish Bose, 1980) and Assam, Chhattisgarh, Jharkhand, Orissa, Rajasthan, and Uttarakhand. These states have reported highest percentage of neonatal deaths. This survey consists of a three round panel survey that interviewed over 4 million households in each round, as well as a one-time Clinical, Anthropometric, and Bio-Chemical Survey (DSDR, 2022). AHS provides data at district level. Census and National Sample Survey Organization (NSSO) also provides data on morbidity or illness or ailing or disability. NSSO provides data for ailing in terms of one before the survey and 15 days before the survey and hospitalization during 365 days before the survey. Census has collected data on disability related to seeing, hearing, speech, locomotor, multiple disorders, mental illness, mental retarded, etc.

1.4 MORTALITY MEASURES

Mortality analysis is one of the most important measures to know the population changes. If we notice, decline of fertility follows death rates. It indicates fertility control is following mortality. Table 1.1 compiles the denominator and numerator and radix for different mortality measures.

Table 1.1: Elements and components of measurement of Mortality

Measure	Numerator	Denominator	Radix
Crude death rate (CDR)	Total number of deaths during a given time interval	Mid-interval population	1,000
Age Specific death Rate	Number of deaths between x and x+n in the year	Number of persons aged between x and x+n at the middle of the year	1000
Cause-specific death rate	Number of deaths assigned to a specific cause during a given time interval	Mid-interval population	100000
Proportionate mortality	Number of deaths assigned to a specific cause during a given time interval	Total number of deaths from all causes during the same time interval	1,000

Death-to-case ratio	Number of deaths assigned to a specific cause during a given time interval	Number of new cases of same disease reported during the same time interval	1000
Neonatal mortality rate (NNMR)	Number of deaths among children < 28 days of age during a given time interval	Number of live births during the same time interval	1,000
Post-neonatal mortality rate (PNMR)	Number of deaths among children 28–364 days of age during a given time interval	Number of live births during the same time interval	1,000
Infant mortality rate (IMR)	Number of deaths among children < 1 year of age during a given time interval	Number of live births during the same time interval	1,000
Maternal mortality rate (MMR)	Number of deaths assigned to pregnancy-related causes during a given time interval	Number of live births during the same time interval	100000

Source: <https://www.cdc.gov/csels/dsepd/ss1978/lesson3/section3.html>

1.4.1 Crude Death Rate (CDR)

Crude Death rate (CDR) is the simplest measure of the general level of mortality in a population. This rate has a simple interpretation, for it gives the number of deaths that occur, on the average, per 1,000 people in the community, or it is the average number of deaths occurring during a year per 1000 persons in the total population. Further, it is relatively easy to compute, requiring only the total population size and the total number of deaths. Besides, it is a probability rate in the true sense of the term. It represents an estimate of the chance of dying for a person belonging to the given population, because the whole population may be supposed to be exposed to the risk of dying of something or the other. However, it has also some serious drawbacks. In using the CDR, we ignore the fact that the chance of dying is not the same for the young and the old or for males and females, and the fact that it may also vary with respect to race, occupation or locality of dwelling.

$$\text{CDR} = \frac{\text{Deaths in a year (D)}}{\text{Population at mid-year (P)}} \times 1000$$

This measure is generally a poor indicator of mortality as it does not take age structure into account. The number of deaths is largely a function of the size and the age-sex structure of the population. The risk of dying in general is high in the infant and childhood ages, a decline drastically thereafter up to age 20 and then rise slowly but steadily, increasing sharply at ages above 50. This type of curve known as the reverse J-shaped curve is characteristics of age patterns of mortality in all populations. Many developing countries with higher mortality levels at every age show a lower than developed countries because the former have a much larger proportion of their population at younger ages due to high fertility. A crucial factor in the study of mortality to take account of its variations by age.

1.4.2 Age Specific Death Rate (ASDR)

The age-cause-specific death rates are obtained by selecting deaths in specific age and cause group of the population as the numerator. It should be noted that the

sum of the cause-specific rates over all causes equals the crude death rate. Similarly, the sum of the age-cause-specific death rates equals the age-specific death rate at a given age. Standardization is a technique, which provides a summary measure of the rates (similar to the crude rates) while controlling for the compositional variation between the populations being compared. Thus, a comparison of the standardized rates gives a 'true' comparison of the phenomenon studied. We shall illustrate the calculations of the standardized rates with the help of the death rates. The ASDR is a type of central death rate, that is, a rate relating to the events in a given category during a year to the average population of the category. In a high mortality situation, the death rates by age, that is, the age specific death rates, form a U-shaped curve indicating a high mortality in early and old ages. At low levels of mortality, the pattern of ASDR changes to J-shaped indicating a relatively higher mortality in the very early period of life, which drops to a low level after the hazards of early life and extends over a long period of life, and finally it rises sharply in old ages.

The age-specific death rates are calculated from deaths and population both specific to each age (or age group) of the population. Thus,

$$\text{ASDR} = \frac{\text{Number of deaths between } x \text{ and } x+n \text{ in the year } ({}_nD_x)}{\text{Number of persons aged between } x \text{ and } x+n \text{ at the middle of the year } ({}_nP_x)} \times 1000$$

Where 'x' indicates the age and 'n' the class interval of age.

1.4.3 Specific Death Rate (SDR)

The crude death rates for specific causes of death are calculated in a similar way by selecting deaths due to specific cause as the numerator and mid-year population as the denominator. Thus,

$$\text{SDR} (m_x) = \frac{\text{Number of deaths to population in age group } x \text{ in an area during the particular year } (D_x)}{\text{Midyear population of age group } x \text{ in the same area during the same year } (P_x)} \times 1000$$

The rates could be made specific to sex by selecting the numerator and the denominator for each sex of the population. This measure is annual age specific death rate while ASDR is the death rate of 5 years age groups.

The pattern of mortality by age is one of the initially high rates in the first year of life (within the first year, mortality rates are highest in the first few days of life, very high in the first month and declines throughout the rest of the year). After the first year the rates decline rapidly to a minimum which usually occurs around the tenth year of life. After reaching this minimum the age specific death rates increase gradually upto about 40 years or 50 years of age and then increases at a relatively rapid rate throughout the rest of life. Generally speaking the decline in the mortality rates in early life is quite abrupt in populations with high mortality.

1.4.4 Maternal Mortality Rate/Ratio (MMR/MMRT)

The maternal mortality ratio (MMR) is widely used type of cause specific mortality rate representing approximately the risk of dying as a result of complications of pregnancy, childbirth and puerperium. This ratio is generally defined as the number

of deaths due to puerperal causes per 100000 live births MMR may be looked upon as an alternative to, or a refined version of the corresponding cause-of-death rate. This rate is defined by the formula-

$$\text{MMR} = \frac{D_p}{B} \times 100000 \quad \text{Or} \quad \text{MMRT} = \frac{D_p}{M} \times 100000$$

where

D_p = Total number of deaths from puerperal causes occurring within 42 days of delivery among the female population aged 15-49 in the given period in the given community.

Puerperal infection occurs when bacteria infect the uterus and surrounding areas after a woman gives birth and hence becomes the cause of death.

B = Total number of live births occurring in the given period in the community.

M = number of married women aged 15 -49 years

Since MMRT is defined per 100000, married women and MMR is defined per 100000 live births, hence MMRT and MMR are connected by the following formula-

$$\text{MMRT} = \text{MMR} * \text{GMFR}$$

The life time risk is defined as the probability that at least one women of reproductive age (15-49 years) will die due to child birth or puerperium assuming that chance of death is uniformly distributed across the entire reproductive span and has been worked out using the following formula:

$$\text{Life Time Risk} = 1 - \left(1 - \frac{\text{Maternal mortality Rate}}{100000} \right)^{35}$$

1.4.6 Incidence Rate (IR)

This rate measures the annual incidence of a disease. The numerator of the rate is the number of times attacks of a certain disease are reported in a year, and the denominator is the population exposed to the disease during the same year. Note that the rate can exceed 1, as there could be repeated attacks of the disease to the same person.

$$\text{IR} = \frac{\text{Number of new cases of specialized disease during a give time period}}{\text{Population at risk (Total population)}} \times 1000$$

1.4.7 Prevalence Rate (PR)

This rate is the proportion of the number of people reported as having a disease at a specific point in time. The numerator is the number of people with a disease and the denominator is the total population.

$$\text{PR} = \frac{\text{Number of persons with diseases in given point of time (Old and New cases)}}{\text{Total Population at same point of time}} \times 1000$$

1.4.8 Infant Mortality Rate (IMR)

The groups of population which are at greater risk of dying are newborn infants and mothers in the reproductive age groups. Conventional infant mortality rate is defined as the number of deaths under age 1 year during specified period divided by the number of live births in the same period and usually expressed per 1000 births.

$$\text{IMR} = \frac{\text{Number of death under 1 year age in a year}}{\text{Number of live births in the same year}} \times 1000$$

IMR is not proper rate but a ratio, as the denominator is not wholly the population at risk of the events in the numerator. Some of the deaths under age 1 year in the given year may be among the births which occurred during the previous year and some of the newborns during the year may die in the next year before reaching their first birthday.

Also the risk of dying the first year of life is not uniform in the interval. The risk is the maximum soon after birth and decreases slowly. During the early weeks, the causes of infant deaths tend to be different from those which occur later. For this reason IMR is often divided into two parts based on the intensity of risk of mortality. The first four weeks or one month, called the neonatal period is found to have a very high risk and the number of infant deaths during this period is termed neonatal mortality rate (NMR).

$$\text{NMR} = \frac{\text{Deaths of babies less than 4 weeks old (<28 Days)}}{\text{Number of live births in the same year}} \times 1000$$

Similarly, death of infants after four weeks is termed post neonatal mortality (PNMR).

$$\text{PNMR} = \frac{\text{Deaths of babies aged 4-52 weeks}}{\text{Number of live births in the same year}} \times 1000$$

1.4.9 Cause Specific Death Rate (CSDR)

For the control of diseases in any population, it is important to know the causes of death. Thus cause specific measures of mortality are useful in understanding the relative contributions of different causes to the overall mortality rates in the population and resources to be mobilized for controlling the specific disease. The most commonly used measure is the cause specific death rate as below-

$$\text{CSDR} = \frac{\text{Number of deaths in a year due to a particular cause (Dc)}}{\text{Midyear total population (P)}} \times 1000$$

Cause can be separately computed by age and sex. It can be also calculated cause specific death ratio.

1.4.10 Child Mortality Rate (CMR)

It is define as the total number of deaths of children aged 1 to 4 (called under 5 mortality) in the given year and geographical region and to the 1000 population of the same age in that year and same geographical regions.

$$\text{CMR} = \frac{\text{Number of deaths children below 5 years of aged in a year in a region}}{\text{Total Population 0-4 years age in a year in a region}} \times 1000$$

1.4.11 Software for Mortality Analysis

There are 15 spreadsheets for analyzing the mortality of a population. Some spreadsheets were prepared for populations with full and acceptable data, while others assume that only limited data are available. Although this set of mortality spreadsheets is the largest, the authors did not include the techniques developed and published by the United Nations in the MORTPAK package. Thus, the analysis of mortality may require the use of both PASEX and MORTPAK packages. The PASEX spreadsheets related to mortality are as follows:

1. **ADJMX**- Adjusts a pattern of mortality rates by age for both sexes combined or for each sex (ASDRs or central death rates from an empirical or existing life table) for obtaining a user-specified number of deaths in a population.
2. **BTHSRV**- Estimates infant mortality rates based on information on the number of children born during a year prior to the census and the number still alive at the census date.
3. **E0LGST**- Fits a logistic function to values of life expectancies at birth for each sex simultaneously, given two or more observed values of life expectancies and the two asymptotes of the logistic.
4. **GRBAL**- Uses the technique developed by Brass for estimating the completeness of reporting of deaths over 5 years of age in relation to a population (Brass, 1975).
5. **and 6. INTPLTM and INTPLTF**- Interpolate male and female life tables, respectively, between the values of two given sets of 'pivotal' life tables.
7. **and 8. OGITQX and LOGITLX**- These two spreadsheets are used for smoothening functions of a life table, using logits of the $q(x)$ and $l(x)$ functions of a life table.
9. **LTMXQXAD**- Constructs a life table from age-specific death rates or from the probabilities of dying between two specific ages.
10. **LTNTH**- Selects a Coale–Demeny model life table, region North, that will reproduce a given crude death rate pertaining to a given population age structure.
11. **LTPOPDTH**- Constructs and smoothen a life table for both sexes or one sex at a time, using population and death data.
12. **LTSTH**- This is the same as LTNTH, but using region South of the Coale–Demeny model life tables.
13. **LTWST**- This is the same as LTNTH, but using region West of the Coale–Demeny model life tables.
14. **PREBEN**- Estimates the level of mortality for ages 5 years and above during an intercensal period using Preston-Bennett method, 1983.
15. **PRECOA**- Uses the technique developed by Preston and Coale for evaluating an available age distribution of deaths in relation to the population.

Check Your Progress 1

Note : a) Write your answer in about 50 words

b) Check your answer with possible answers given at the end of the unit.

1) Explain Mortality and Morbidity

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2) List number of mortality measures and defined IMR

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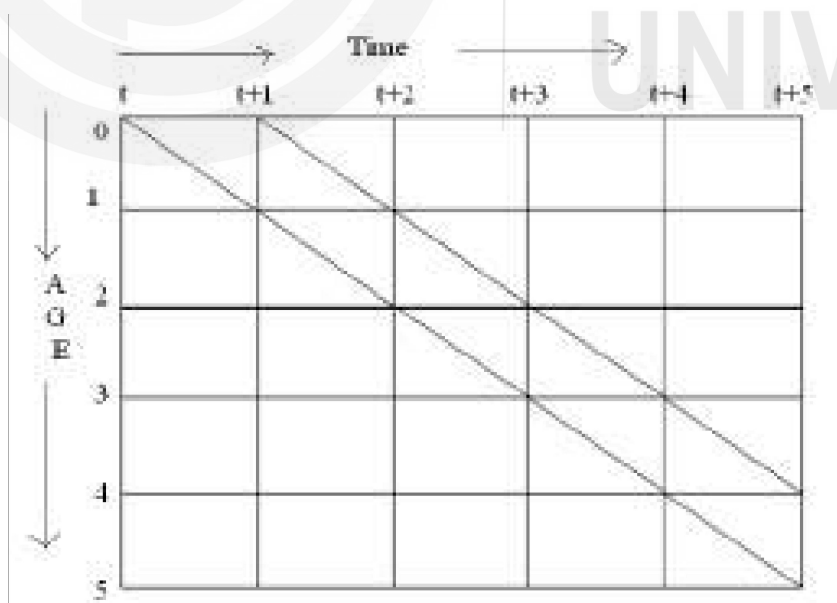
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1.5 LEXIS DIAGRAM

Lexis diagram is a two dimensional diagram used to characterize events that occur to individuals belonging to different cohorts. In this diagram time is considered along the horizontal axis and age along vertical axis. Each individual member of a population is represented by a line at 45° of either axis starting at age $x = 0$ and at the moment of birth and terminating at a point which corresponds to both his age and time of death. A part of diagram can be represented as follows:

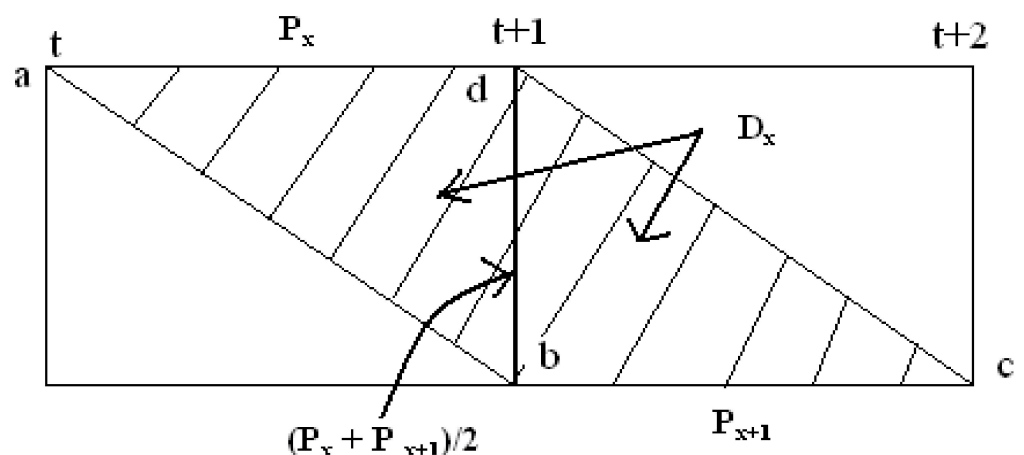


Source : Srinivasan, 1998

In the Lexis diagram, the time of occurrence of event is shown on the X- axis and the duration since an initial reference data (e.g. Age since birth) on the Y-axis. In the diagram above, time refers to the exact time of occurrence, and age, the exact age of the person.

1.6 COHORT RATES

The rates for the cohorts are usually based on the probability concept. However, cross-section rates for the cohorts (Known as Central rates) are also calculated. Consider the following Lexis diagram.



Source : Srinivasan, 1998

If we know the population at exact age X during time t- t+1 (say P_X) and at exact age X+1 (say P_{X+1}) during time t+1- t+2, and the deaths occurring to this cohort in the parallelogram abcd (say D_X), The probability of dying between exact ages X and X+1 will be $= D_X / P$

The central death rate at age X will be =

$$\frac{D_x}{P_x + P_{x+1}/2}$$

$(P_x + P_{x+1})/2$ will be the average or mid - year population (represented by the vertical line 'db') on the assumption that the deaths have occurred uniformly over the parallelogram 'abcd'. However, this is not true for deaths, which occur at very early ages, especially among the infants in their first 12 months of life. The mid-year population is estimated by the use of the separation factors.

1.7 STANDARDIZED DEATH RATE

1.7.1 Method of Direct Standardization

In this method the distributions of the compositional variables (age, sex, marital status etc.) of the populations that are being compared, are made identical and the standardized rates (similar to the crude rates) are calculated such that the difference between them is only due to the variation in the age-specific rates of their population.

A Standard population is selected which is employed for deriving all the standardized rates in a set to be compared.

Data Needed

(1) For one compositional variable (say age) standardization, age distribution of the standard population, and

(2) Age-specific death rates in all populations to be compared.

Calculations

A. If $M(i, x)$ represents the age-specific death rate at age (i) for population (x), and $P(i, s)$ is the standard population at age (i), $P(s)$ is the total standard population. The standardized death rate for population ' x '

$$\sum \frac{P(i, s)}{P(s)} M(i, x)$$

The numerator is the number of expected deaths in the standard population had the age-specific death rates of population (x) applied to the standard population, and the denominator is the total standard population. The rate is multiplied by 1,000 to express the rate as per 1,000 populations. (All the calculations are done with the rates per person. Finally, the standardized death rate is multiplied by the constant 1,000).

B. If the standardized death rate is required after controlling for the two characteristics of the population, say age and sex, the data needed will be the same as on the previous page but split by sex as well. Thus, the standardized death rate for population x will be:

$$\frac{\sum_i [P(i, s, \text{males}) * M(i, \text{males}) + P(i, s, \text{females}) * M(i, \text{females})]}{[P(s, \text{male}) + P(s, \text{female})]}$$

C. If the death rates of males and females are to be compared, these are two different populations, and the method given under A is to be used. Thus,

$$\text{The standardized death rate of males} = \frac{\sum_i [P(i, s) * M(i, \text{males})]}{P(s)}$$

$$\text{The standardized death rate of females} = \frac{\sum_i [P(i, s) * M(i, \text{females})]}{P(s)}$$

The Standard Population

The selection of the standard population is, in theory, arbitrary. However, this population should be similar to the ones for whom the rates are being compared. The population of India at a most recent census data is appropriate for measuring state differentials in mortality, or for comparing mortality trends over time for India. If two country's rates are to be compared either one country's age distribution or the average of the two country's distributions is appropriate.

1.7.2 Method of Indirect Standardization

This method is used when the age-specific death rates for the populations to be compared cannot be calculated because of the distribution of the number of death by age is unavailable or not reliably available, but the total number of deaths and the age distribution of populations whose rates are to be compared are available.

Data Needed

- (1) Observed number of deaths in all populations whose death rates are to be compared.
- (2) Age distribution of all populations whose death rates are to be compared.
- (3) Age-specific death rates for a population to be used as standard.
- (4) Crude death rate in the standard population.

Calculations

A. If $P(i,x)$ represents the population 'x' at age (i), $M(i,s)$ is age-specific death rate at age 'i' in standard population, $M(s)$ is the crude death rate in the standard population, $O(x)$ is the observed number of deaths in population 'x', and $E(x)$ is the expected deaths in population 'x', then

$$\text{The standardized death rate for population } x = \frac{O(x) * M(s)}{E(x)}$$

$$\text{The expected deaths in population } x = E(x) = \frac{\sum_i P(i,x) * M(i,s)}{1,000}$$

(All the calculations are done with the rates per person. Finally, the standardized death rate is multiplied by the constant 1,000).

Foetal Death: It is known as the death prior to the complete expulsion or extraction from its mother of a product of conception at any point of time of pregnancy.

Still Birth: It is used for death of foetus after completing 28 weeks and till the time of birth.

1.8 MEASURE OF MORBIDITY

The aim of morbidity study is to review some of the problems and techniques that have arisen in the different mode of measurement of sickness applicable to different definitions of ill health and to different agencies identifying ill health (IIPS Study material MPS05, B1U1). Morbidity measures have some duration of the study such as beginning period to ending period of illness or ailing. Thus, four reference periods are-

- i. illness beginning during the period and ending during the period
- ii. illnesses beginning during the period and still existing at the end of the period.
- iii. Illness existing before the beginning of the period and ending during the period.
- iv. Illnesses existing before the beginning of the period and existing at the end of the period.

1.8.1 Crude Incidence Rate (CIR)

Crude incidence rate is defined by the number of new cases of illness or injury (I)

in a community on an annual basis per 1000 average number of persons (P) living in the community during the reference period.

$$\text{CIR} = \frac{I}{P} \times 1000$$

In the case of acute illnesses of a relatively short duration, the period prevalence rate is approximately equal to the incidence rate when the period of observation is sufficiently long.

1.8.2 Attack Rate

When a population is exposed to the risk of a disease for a limited period of time only, the incidence rate may be termed as the attack rate. If this disease found in any of the family members this person known as Index person

$$\text{Secondary Attack rate} = \frac{\text{The number of new illnesses with the same disease of the index person}}{\text{Number of persons exposed excluding the index person}} \times 1000$$

1.8.3 Prevalence Rate (PR)

Prevalence of a disease indicates how common the disease is in a population. It is of two types- period (PPR) and point prevalence rate (PoPR). PPR is used to measure the frequency of illness in existence during a defined period (day, month, year). While PoPR is used to measure the number of cases of illness, new and old, existing at a particular point of time such as 9 AM on Monday, the 27 September 2022 and related to the number of persons exposed to the risk at that point of time. PoPR is more commonly used to know the case is new or old.

The prevalence rates are determined for chronic illnesses such as leprosy, tuberculosis, filariasis, cancer, diabetes, HIV/AIDS, etc., where the onset is very ill defined. Incidence rate are usually calculated for diseases whose onsets are clearly known such as malaria, cholera, measles, etc.

$$\text{Average duration of illness per person} = \frac{\text{Total days of illness of sick persons}}{\text{Total population exposed}}$$

$$\text{Average duration of illness per person sick} = \frac{\text{Total days of illness of all sick persons}}{\text{Number of person fell ill}}$$

$$\text{Average duration of illness per spell} = \frac{\text{Total days of illness of all sick persons}}{\text{Number of spell of illness}}$$

1.8.4 Case Fatality Rate (CFR)

CFR is intended to measure the risk of death from a specific condition among those suffering from it. It is usually computed only for acute conditions of a relatively short duration. For accuracy the deaths D_i should be related only to the causes from which they may arise.

$$\text{CFR} = \frac{\text{Di (number of deaths from cause i) during specified period}}{\text{Ii (number of new cases of illness or any cause i) same period}} \times 1000$$

$$\text{Cure Rate} = \frac{\text{The number of cases still alive and without evidence of the disease 5 years after treatment}}{\text{Number of cases treated during the reference period under study}}$$

1.8.5 Mutually Inclusive Measures

P=Prevalence, I=Incidence, D= Duration, M= Mortality, F= Fatality

$$P = I \times D$$

$$F = M / I$$

Check Your Progress 2

Note : a) Write your answer in about 50 words

b) Check your answer with possible answers given at the end of the unit.

1) What is standard population?

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2) What is reference period for Morbidity measure?

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1.9 LET US SUM UP

You have studied mortality concepts and mortality measures, with specific data for mortality and morbidity measures such as SRS, CRS and AHS along with NSSO data for morbidity. Different measures of mortality such as CDR, ASDR, MMR, IMR, CDR, etc. have read in this unit followed by morbidity measures such as incidence rate, prevalence rate, cure rate. etc. you have also studies on interrelationship between morbidity and mortality measures. Ailing and illness is the health conditions of person in the households during a reference period. In next unit you will be read the patterns of mortality across the states in India and at international by age and sex.

Acknowledge

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1.11 CHECK YOUR PROGRESS - POSSIBLE ANSWERS

Check Your Progress 1

Q1. Explain Mortality and Morbidity

- A. Mortality means number of deaths occurs during a specific period of time which may be natural deaths, age specific death, accidental death, etc. while

morbidity means the state of health of the person who has felt sickness during a reference period. The morbidity may be due to communicable disease or non communicable diseases. Morbidity may be cause of death but all mortality is not due to morbidity.

Q2. List number of mortality measures and defined IMR

- A. There are number of mortality measures such as crude death rate, specific death rate, infant mortality rate (IMR), maternal mortality rate, child death rate, etc. IMR is the best indicator of reproductive and child health care system of any country. IMR is defined as number of child deaths below one year age over per 1000 live births taken place during same year.

Check Your Progress 2

Q1. What is standard population?

- A. The standard population should be similar to the ones for whom the rates are being compared. The population of India at a most recent census date is appropriate for measuring state differentials in mortality, or for comparing mortality trends over time for India. If two country's rates are to be compared either one country's age distribution or the average of the two country's distributions is appropriate.

Q2. What is reference period for Morbidity measure?

- A. reference period is important for morbidity measures, which have some duration of the study such as beginning period to ending period of illness or ailing. There are four reference periods for morbidity measure -
1. Illness beginning during the period and ending during the period
 2. Illnesses beginning during the period and still existing at the end of the period.
 3. Illness existing before the beginning of the period and ending during the period.
 4. Illnesses existing before the beginning of the period and existing at the end of the period.